EMAT101L: Engineering Calculus (Odd Semester 2023-24) Tutorial 5 solution hints

Mark as done

Tutorial 5 Solution

① Let $a_n = \frac{n+1}{n^p}$ & $b_n = \frac{1}{n^{p-1}}$ then $\lim_{n\to\infty} \frac{a_n}{b_n} = \lim \frac{n+1}{n} = 1 \neq 0$ finite

By limit comparison test, $\sum a_n$ & $\sum b_n$ converge or diverge together

Now $\sum b_n = \sum \frac{1}{n^{p-1}}$ Cov. if $p-1 > 1 \Rightarrow p > 2$

& div if $p-1 \leq 1 \Rightarrow p \leq 2$.

② $a_n = \frac{x^{n+1}}{(n+1)\sqrt{n}}$, $a_{n+1} = \frac{x^{n+2}}{(n+2)\sqrt{n+1}}$

$\therefore \lim_{n\to\infty} \frac{a_{n+1}}{a_n} = x \lim_{n\to\infty} \frac{n+1}{n+2}\sqrt{\frac{n+1}{n}} = x$ By Ratio test, $\sum a_n$ con if $x < 1$ & div if $x > 1$

At $x=1$, $a_n = \frac{1}{(n+1)\sqrt{n}}$ Take $b_n = \frac{1}{n\sqrt{n}}$

$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{n}{n+1} = 1 \neq 0$ finite $\therefore$ By limit comparison test $\sum a_n$ & $\sum b_n$ conv. or div. together

$\because$ by p-series test, $\sum b_n = \frac{1}{n^{3/2}}$ conv.

$\therefore \sum a_n$ also conv. at $x=1$ also.

③ Let, if possible, the given series be cgt.

By Leibnitz Test, $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt{n}}$ is cgt $\because \lim a_n = \lim \frac{1}{\sqrt{n}} = 0$

& $a_n = \frac{1}{\sqrt{n}}$ is mono. $\downarrow$ing.

Thus, $\sum_{n=1}^{\infty} \left[\left(\frac{1}{n} + \frac{(-1)^{n-1}}{\sqrt{n}}\right) - \frac{(-1)^{n-1}}{\sqrt{n}}\right]$ is cgt

$\Rightarrow \sum \frac{1}{n}$ is cgt which is false. Hence the given series is not cgt.

◀ Tutorial 4 solution hints